

ANN6: Forward & Backpropagation – Detailed Code & Viva Explanation

Cell 1: Import Library

Code:

```
import numpy as np
```

Explanation (what to say in viva):

Here I am importing numpy library.

It is used for matrix operations required in neural networks.

Backpropagation involves heavy mathematical computations.

Cell 2: Initialize Inputs and Outputs

Code:

```
x = np.array([[0,0],[0,1],[1,0],[1,1]])  
y = np.array([[0],[1],[1],[0]])
```

Explanation (what to say in viva):

X contains input combinations.

y contains expected outputs (XOR problem).

This is supervised learning.

We train the network using known outputs.

Cell 3: Initialize Weights

Code:

```
w1 = np.random.rand(2,2)  
w2 = np.random.rand(2,1)
```

Explanation (what to say in viva):

W1 is weight matrix for input to hidden layer.

W2 is weight matrix for hidden to output layer.

Random initialization helps learning.

If weights are zero, learning will not happen.

Cell 4: Activation Function

Code:

```
def sigmoid(x):  
    return 1/(1+np.exp(-x))
```

Explanation (what to say in viva):

Sigmoid converts values between 0 and 1.

It introduces non-linearity.

Used in hidden and output layers.

Cell 5: Forward Pass

Code:

```
z1 = np.dot(X, W1)
a1 = sigmoid(z1)
z2 = np.dot(a1, W2)
a2 = sigmoid(z2)
```

Explanation (what to say in viva):

z1 is input to hidden layer.

a1 is activation of hidden layer.

z2 is input to output layer.

a2 is final output.

This process is called forward propagation.

Cell 6: Error Calculation

Code:

```
error = y - a2
```

Explanation (what to say in viva):

Error is difference between actual and predicted output.

It tells how far prediction is from correct value.

This is used to update weights.

Cell 7: Backpropagation

Code:

```
dw2 = np.dot(a1.T, error)
dw1 = np.dot(X.T, error)
```

Explanation (what to say in viva):

Gradients are calculated for weight updates.

These gradients indicate direction of improvement.

This step propagates error backward.

Cell 8: Update Weights

Code:

```
W1 += dw1
W2 += dw2
```

Explanation (what to say in viva):

We update weights using gradients.

This improves model performance.

Repeated updates reduce error.

Cell 9: Output

Code:

```
print(a2)
```

Explanation (what to say in viva):

This prints final output.

It should be close to expected output after training.

This shows learning success.